



The role of experience in shifting gender beliefs on performance: Experimental evidence[☆]

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ABSTRACT

In labour markets, women are often underrepresented relative to men. This underrepresentation may be due to inaccurate beliefs about ability across genders. Inaccurate beliefs might cause a sampling problem: to have accurate beliefs about a group, one must first collect information about it. However, employers may not wish to shortlist individuals from a group that is perceived to exhibit lower quality. Inaccurate beliefs may also persist due to biased belief updating. We run a stylised hiring experiment to disentangle these two effects. We ask participants to create shortlists from a male and a female pool of workers and give them feedback on the skill of those they shortlist. Based on that information, participants hire workers, and provide us with their beliefs about the distribution of skills in the male and female pools. We study how employers update their beliefs as a function of their past shortlisting behaviour, and how they shortlist given their beliefs. Participants were more likely to sample from the pool with the higher subjective mean quality (on average men) and lower subjective variance. Participants were not Bayesian updaters but there were no gender-specific biases in updating. Sampling more from a pool and spending more time sampling yield more accurate beliefs.

1. Introduction

Women are often underrepresented relative to men in many domains (Reuben et al., 2014; Cheryan et al., 2017). In UK politics, 35% of House of Commons MPs and 28% of the members of the House of Lords are female (Uberoi and Mansfield, 2022). In UK labour markets, the pattern is similar: in business, 37.7% of directors of FTSE 100 companies are women; in law enforcement, 32% of police officers are female; in academia, only 28% of UK professorships are held by women (Buchanan et al., 2022; HESA, 2022). This under-representation has serious consequences for economic performance and growth, as institutions are under-recruiting from roughly 50% of the talent pool.

A prominent explanation for discrimination in labour markets is statistical discrimination, in which employers use statistical information pertaining to observable characteristics to help them make judgements about individuals in the absence of information about their productivity (Phelps, 1972; Arrow, 1977).¹ Recent work in this literature explores the role of inaccurate beliefs in

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statistical discrimination, in which employers do not have access to an unbiased signal about productivity, and instead rely on (potentially inaccurate) subjective beliefs about a group (Bordalo et al., 2016; Bohren et al., 2019, 2020; Feld et al., 2022).

Inaccurate statistical discrimination can create a sampling problem: to form accurate beliefs about a group, employers must engage with its members, but information is costly to acquire (Bartoš et al., 2016; Fershtman and Pavan, 2021). If employers hold negative priors about a group, they may see little value in recruiting from it, thereby reinforcing inaccurate beliefs over time (Lepage, 2023). We focus on two mechanisms that drive persistent belief gaps about performance in labour markets. The first is biased sampling: employers may search more among groups they perceive as higher quality, resulting in more accurate beliefs about those groups. The second is biased belief updating: employers may interpret signals differently depending on the group, potentially due to motivated reasoning, leading them to reinforce existing stereotypes (Eil and Rao, 2011; Möbius et al., 2022; Coffman, 2014; Bordalo et al., 2016, 2019).

In this paper we present evidence from a laboratory experiment that examines these mechanisms while controlling for priors. We are interested in studying information search and belief updating about performance across genders in a highly stereotyped context. As such, we run a stylised hiring task paired with a belief updating task. Our participants took on the role of employers. At the start of the experiment, we elicited employers' prior beliefs about the distributional performance of an all-female pool of candidates and an all-male pool of candidates. Employers did not have any information about the performance distribution in either pool. Thereafter, in each round of the experiment, employers constructed a shortlist of 10 individuals. In particular, they chose how many shortlisted individuals would be drawn from the all-female pool and how many would be drawn from the all-male pool. Employers then found out the productivity of each shortlisted candidate and selected one. At the end of each round, we elicited the employers' revised beliefs about the distribution of performance for each of the gender-specific pools.

Employer payoffs were proportional to their selected candidate's performance in a real effort task: the number of chess puzzles successfully solved in 5 min (based on the performance data of Chess.com's "Puzzle Rush" task). We chose chess as the domain for our study because it is a strongly male-stereotyped activity, making it a suitable setting to investigate belief-based discrimination. Chess remains male-dominated both in participation and cultural perception. For instance, women account for only 11% of FIDE-rated players at all levels, 1% of the top-100 players, and only 2% of Grandmasters (Smerdon, 2022). These disparities are often reinforced by public narratives. Nigel Short, a grandmaster and former vice-president of FIDE, has publicly claimed that men are "hardwired" to be better at chess than women (Ellis-Petersen, 2015).

However, these gender differences in performance need not be innate. Chess ability is a function of an individual's capacity to do backward induction, and working memory (Robbins et al., 1996). There is no evidence of gender differences in these cognitive functions (Robert and Savoie, 2006; Harness et al., 2008). Performance differences across gender in chess may be instead the result of social, structural or psychological factors, such as exposure to the game, perseverance, competitiveness, or stereotype threat (Blanch et al., 2015; Blanch, 2016; Maass et al., 2008; Niederle and Vesterlund, 2007; Rothgerber and Wolsiefer, 2014; Smerdon et al., 2020). Focusing on puzzles instead of actual games removes two dimensions which may be at the heart of gender differences in performance: competitiveness (Niederle and Vesterlund, 2007), and risk or ambiguity attitudes (Borghans et al., 2009).²

We find significant gender differences in the employers' belief distributions about the performance of men and women. In particular, employers initially believe the all-male pool had a higher average performance than the all-female pool. However, there is difference in priors about the variance in performance between the two gendered pools. The difference in beliefs about mean performance is driven by differences in extremes: employers initially believe the all-male pool to have more high-performers and fewer low-performers.

We then study whether these beliefs persist, and the role endogenous information search and biased belief updating play in determining that persistence. We find no evidence for biased belief updating. When provided with new information about the productivity of male or female workers, employers do not update their beliefs differently for either gender.

We do find evidence for endogenous information search: employers sample more from the pool that they believe has a higher mean and/or a lower variance in performance. Additionally, employers' beliefs about a group get more accurate as they obtain more information about that group. These two mutually reinforcing results provide support to the information search mechanism we conjectured. Finally, we find soft forms of affirmative action may not elicit a change in sampling behaviour, and as such, provide no stimulus to correct inaccurate beliefs.

We intentionally constructed the two gendered pools of candidates such that, unbeknownst to participants, they had very similar distributions of performance, since we are interested in understanding how inaccurate beliefs can persist about similarly performing groups, and what role belief updating and sampling play in determining that persistence. Our experimental environment might not match real-world performance distributions. In the real world, men might perform better on average than women in chess puzzles, perhaps due to self-selection. This would be consistent with observed gender disparities in competitive chess more broadly. Therefore, any stereotypes driving the initial beliefs of our participants that women perform better in chess puzzles could be accurate. In fact, subjects even might underestimate the extent of this real world difference in our setting. Chess.com does not collect gender information from its users, so we cannot obtain representative gender data on chess puzzle performance. As such, we cannot compare the beliefs elicited in our experiment and actual gender performance in this task.

¹ The other main theory of discrimination is taste-based discrimination (Becker, 1957), which can be described as having a disutility or utility from interacting with members of certain groups, creating a preference to either hire or avoid hiring from those particular groups. See Riach and Rich (2002), Bertrand and Dufló (2017), Blau and Kahn (2017) and Neumark (2018) for reviews of the literature on discrimination in labour markets.

² A caveat that risk attitudes are context-dependent is warranted here (Weber et al., 2002).

Table 1
Summary statistics of performance in both pools.

	Mean	Var	Min	Max
Men/'Pot A'	18.33	34.89	7.00	28.9
Women/'Pot B'	18.29	34.81	7.50	27.00

Our results contribute to the literature on inaccurate statistical discrimination (Bordalo et al., 2016; Bohren et al., 2019, 2020; Feld et al., 2022). Work in this area has focused on how inaccurate priors might persist, and how one may correct them (Komiya and Noda, 2020; Lepage, 2023). We provide experimental evidence on how soft affirmative action models, through the imposition of shortlisting quotas, can impact belief updating. We also contribute to the literature focusing on how past experience with members of a group drives decisions to hire from that group. Using data from a large US retailer, Benson and Lepage (2023) find that a store manager's decision to hire from a group is strongly affected by their own past experiences with members of that group. Sarsons (2017) uses data on physicians' referrals to surgeons and finds that physicians respond differently to both good and bad signals about male and female surgeons. We expand on this literature by providing evidence as to how beliefs about a group change in response to experience and how those changes in beliefs in turn drive information acquisition through shortlisting.

The rest of the paper proceeds as follows. Section 2 presents the experimental design. Section 3 presents the results. Section 4 discusses these results. Section 5 concludes.

2. Experimental design and procedures

As per previous stylised hiring experiments, our participants act as 'employers'. In each experimental round, employers had to (i) state their beliefs about the distribution of performance in an all-male and an all-female pool of workers; (ii) construct a shortlist of 10 candidates drawn from either pool in any combination; (iii) select one candidate after observing the true performance of the shortlisted candidates; (iv) revise their beliefs about each pool. The payoff to employers was a function of the performance by their chosen candidate in a real effort task.

2.1. The real-effort task and the worker pools

The two pools of workers were drawn from publicly available information on profiles on chess.com where players have completed '5 min puzzle rush', a timed chess puzzle challenge on chess.com. In it, players have five minutes to complete as many puzzles as possible. Puzzles increase in difficulty as the player progresses through the challenge. The challenge ends if time runs out or if the player answers three puzzles incorrectly.

Across the publicly available profiles on chess.com, we collected profiles of both genders. We sampled individuals with similar ability as per their performance rating on the website. We classified these profiles by gender based on a publicly available profile picture, a publicly available username, or a chess title.³ We excluded any profile that did not have any recorded attempts at the 5 min puzzle rush, as well as any profile we could not classify with a gender. We calculated the average performance score for each eligible profile based on their recent history of play. This score acts as our performance measure within the main experiment. Across all individuals who attempted 5 min puzzle rush at least once, we created two pots of 50 individuals. One pot had only male players, while the other had only female players.

We constructed these pools such that they had similar performance distributions (see Table 1 for summary statistics on both pools and Fig. 1 for a histogram) and the same number of workers in a performance range. For example, both pools had five workers who had solved between 5 and 10 puzzles. We note that the distribution of performance in either pool does not necessarily reflect real-world performance distributions, which are not known. In practice, men might perform better on average than women in chess puzzles in line with broader gender disparities in competitive chess. While our design enables clean inference on belief dynamics about similarly performing groups, it does not speak to the real-world accuracy of elicited beliefs.

Using chess puzzles rules out two behavioural factors that may rationalise differences in performance across genders in actual chess matches: risk attitudes and competitiveness (Niederle and Vesterlund, 2007; Gerdes and Gränsmark, 2010; Smerdon, 2022). It is possible that by removing the competitive element, we are removing a driver of the male stereotyping of chess. This may limit the size of the effect concerning the beliefs about the differences in performance. Chess puzzles are fundamentally a test of cognitive ability. Yet, gender stereotypes about chess ability may persist in this setting for two main reasons. Firstly, most of our participants are not active players on chess.com and thus lack direct exposure to the specific performance distributions for chess puzzles on the platform. Secondly, learning about the true objective distribution of performance on chess.com is costly and requires substantial time and engagement, which our participants are unlikely to have undertaken prior to the experiment. Given this limited exposure to chess puzzles specifically, and the inability to assess the entire distribution, participants may rely on their existing beliefs about overall chess performance as a proxy. Therefore, even in a context focused solely on cognitive ability, we anticipate that participants might default to using their beliefs about general chess ability, leading to the persistence of gender stereotypes.

³ A player that holds the title of Female Master (FM) or Female Grandmaster (FGM) is necessarily female.

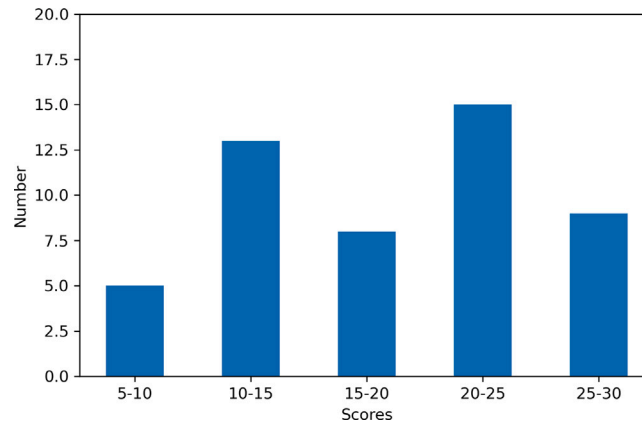


Fig. 1. Distribution of performance in the worker pools. Note: Both pools have the same number of workers in each performance interval.

2.2. An experimental round

The experiment consisted of six rounds, in which participants had to do several tasks in sequence. We explain each task in turn.

2.2.1. Belief elicitation task

At the start of each round, we elicited participants' full distribution of beliefs about the performance of the individuals in each pool. That is, instead of eliciting the first moment of their belief distribution (i.e., the mean, median or mode), as is common in the literature, we elicited the full distribution.

The belief elicitation mechanism worked as follows. For each pool, we presented the participant with a set of 5 sliders, each referring to a discrete set of intervals of possible performance levels: [5,10), [10, 15), [15, 20), [20, 25), and [25, 30). The lowest performance across the pools was 7 and the highest score across all pools in all sets was 28.9. Participants used the sliders to indicate their belief about the relative frequency of each performance interval in a given pool, which corresponded to the probability of drawing an individual whose performance was in that range. The stated probabilities in all 5 sliders had to sum to 100%.

In the first round, all sliders were set to 0. In all subsequent rounds, the starting position of the sliders was the belief distribution given by the participant in the previous round.

In the first round, we incentivised beliefs through a generalised quadratic scoring rule (Harrison et al., 2017). We implemented this scoring rule as follows. For each performance interval, a payoff was associated with that interval, based on how many individuals that the participant believed got a score in that interval. The payoff for an interval was:

$$S = 1.5 + 1.5 \left[2 \times r_k - \sum_{i=1}^5 r_i^2 \right] \quad (1)$$

where r_k is the number of individuals assigned to an interval.

At the end of the experiment, a worker was randomly drawn from one of the two pools. The employer was then paid based on the interval in which the randomly drawn worker's score fell. As employers adjusted the sliders for an interval, they could see in real time the payoff they would receive should the random worker be drawn from that interval. This tool had the advantage of not requiring an explanation about the underlying formula that determined their payoff.

In all subsequent rounds, we did not incentivise beliefs. The only change in how the sliders were presented was that the payment values next to the sliders were removed. We chose to only elicit incentivised beliefs in the first round of the experiment, prior to any hiring decision, because incentivising beliefs in all rounds could interfere with the hiring task due to incentive compatibility concerns. By shortlisting from a pool, an employer acquires information about that pool; as such, incentivising the belief elicitation in later rounds might have artificially led employers to sample more from a particular group (to learn more about it) than they otherwise would have done. Incentivising beliefs in the first round provides a good compromise by ensuring accuracy at the outset, while minimising distortions due to this exploration in later rounds. While we acknowledge that incentivising beliefs is generally preferable, we believe the lack of incentives in later rounds is not a significant problem, as there is evidence that introspection may work sufficiently well (Charness et al., 2021).

2.2.2. Worker shortlisting and hiring task

After beliefs were elicited, participants had to create a shortlist of 10 workers. Participants first stated how many workers they wished to sample from each of the two pools; they could state any number between 0 and 10 for each pool, subject to the constraint that both numbers had to add up to 10. Participants made this decision for the pools simultaneously. The software then randomly selected the specified number of workers from either pool, and created the shortlist.

The employer then saw the profiles of the drawn candidates. Each profile contained that worker's actual performance and the pool from which they were drawn. The employer then had to choose one worker from the shortlist. While this step is somewhat redundant, we keep it for completeness. Employers earned a payoff equal to the chosen worker's score times £0.05, rounded down.

2.2.3. End of round

Prior to the start of data collection, we pre-drew a sequence of random numbers from 1 to 20. At the end of each round, participants were presented with a number from this sequence. Participants were told if the drawn number was 18 or less, they would continue to the next round, else the experiment would end. This pre-drawn sequence was used for all participants, ensuring that the game concluded after the 6th round for everyone. This approach allowed us to maintain control over the experiment's length, especially given that it was conducted online.

After the sixth round was completed, we elicited participants' individual risk aversion using the bomb risk elicitation task (BRET; Crosetto and Filippin, 2013).

Participants were paid for three events. Firstly, they were paid for the outcome of the belief elicitation task in the first round. Secondly, they were paid for the performance of the chosen worker in the final round, at a rate of £0.05 multiplied by the selected worker's score, rounded down. Thirdly, they were paid for their decisions in BRET at a rate of £0.03 for each box opened. Participants also received £3 for completing the experiment.

The average payoff was £2.90 plus a show up fee of £3, for an experiment which lasted on average 25 min. Our experimental software was programmed in oTree (Chen et al., 2016). We recruited 600 participants through Prolific.co. Among the participants, 50.8% were female, 15% were students, and 44.6% were employed full-time. The average age was 37.5 years. Nearly half of the participants (49.5%) had experience in a hiring role. Regarding chess-related experience, 65% of participants were familiar with chess rules. Only 4.7% of participants had played Puzzle Rush. Additionally, 35.5% of the participants had watched 'The Queen's Gambit'.

The experiment (methods, sampling, procedures and statistical analysis) was pre-registered on OSF (<https://osf.io/8n5sq/>).⁴ All statistical analysis reported in the paper is pre-registered unless signposted as exploratory.

2.3. Hypotheses

Our first hypothesis concerns prior beliefs. As we argued earlier, chess is a male-dominated activity, in which there are ingrained gender stereotypes about performance. It is possible that women are perceived to have higher variance in ability, or that the beliefs about women are noisier and result in higher variance (Aigner and Cain, 1977). As a result, we hypothesise that these stereotypes will carry over to our experimental setting. Notably, this contrasts with the Greater Variability Hypothesis, which suggests that male ability is more variable than female ability, particularly at the extremes of performance (Howard, 2005).

Hypothesis 1a (*Mean of Prior Beliefs*). Employers' beliefs about the performance of male workers will have a higher mean than beliefs about performance of female workers.

Hypothesis 1b (*Variance of Prior Beliefs*). Employers' beliefs about the performance of male workers will have a lower variance than beliefs about performance of female workers.

Our second hypothesis pertains to how employers update their beliefs conditional on new information. Gender bias in performance could manifest itself in how individuals update their beliefs given new information about male and female performance. It may be that stereotypes drive gender difference and as such participants update their beliefs in a direction that would match these gender stereotypes (Bordalo et al., 2019). Additionally, it may be that participants wish to hold these gender stereotyped beliefs, further encouraging belief updating in this direction through motivated reasoning (Eil and Rao, 2011; Möbius et al., 2022; Thaler, 2020).

Hypothesis 2 (*Belief Updating*). Employers of all genders will update their beliefs differently for women than for men.

Beliefs may persist through belief-driven sampling behaviour: if an individual only samples from their a priori better-perceived group, they will never hire from their worse-perceived group. Under Bayesian inference, when an individual draws a very large sample, their prior beliefs are crowded out by the observed data. Under the assumption that a large sample is more likely to be reflective of the true population, an individual's posterior belief will be more in line with this representative sample. In our case, a larger sample occurs with a greater number of observations being drawn across the rounds played. As such, with a greater number of observations, an individual's posterior belief will converge to the true distribution, which is analogous to saying that the participant will have a greater level of accuracy in their beliefs relative to the true distribution of performance.

Hypothesis 3 (*Sampling and Belief Accuracy*). The more sampling from a given pool, the more accurate beliefs will be about the ability of workers in that pool.

Given the previous discussion about the relationship between the accuracy of beliefs and sampling, any difference in beliefs about similar groups may be due to the fact that individuals have not sampled sufficiently from one or both groups. One policy solution to ameliorate this problem is the use of some form of soft affirmative action in shortlisting, requiring employers to shortlist a minimum number of candidates from a specific group. Under such a rule, previously under-sampled groups will be sampled more, and greater sampling will lead to more accurate beliefs.

⁴ Deviations from the pre-registration can be found in Appendix A.1.

Hypothesis 4 (Sampling Restrictions and Belief Accuracy). Shortlisting restrictions on the number of workers an employer can shortlist from a given gender will improve the accuracy of beliefs about both groups.

We conclude by looking at shortlisting. A risk neutral employer should only care about the mean of the distribution of performance, and shortlist from the pool with the highest mean. However, for risk averse employers, all else equal, a distribution of performance with a higher variance will be less appealing, as the downside utility risk will outweigh (in utility terms) the upside risk.

Hypothesis 5 (Sampling Behaviour). Employers are more likely to sample from a candidate pool the higher their mean beliefs about performance and the lower the variance in their beliefs about that pool.

2.4. Experimental design

2.4.1. Main treatments

We are primarily interested in how participants update their beliefs and how they sample workers. For these purposes, we have three main treatments. The first is our baseline treatment.

Baseline (BASE): Employers could sample from the pools in any combination they preferred, ranging from 10 men and 0 women to 0 men and 10 women. We only incentivised the first round of beliefs, in order not to induce over-sampling of a group to obtain more accurate beliefs about that group. However, we would still like to understand how participants update their beliefs when giving accurate beliefs is incentivised. The Only Beliefs-Framing treatment (OB-F) addresses this question.

Only Beliefs-Framing (OB-F): Employers only completed the belief elicitation task and did not shortlist workers. At the start of each round, participants provided their beliefs about each worker pool. We then drew a random number of workers from each pool. These profiles, including each worker's score and the pool they came from, were presented to employers, but employers made no hiring decision. Unlike the treatments with a shortlisting task, employers' payoffs were determined by the beliefs given in the final round, using the generalised quadratic scoring rule. However, since the number of rounds was unknown to participants, it was always in their best interest to treat each round as payoff-relevant. As such, all beliefs in this task were effectively incentivised.

Shortlisting Restrictions (SR): Employers were required to draw at least one profile from each pool at the stage of creating a shortlist. The SR treatment provides a test of [Hypothesis 4](#) and the idea that restrictions on shortlisting may improve the accuracy of beliefs by enforcing a minimum amount of shortlisting from each pool. This treatment removes the possibility that a participant never learns about one of the worker groups and as such, provides a route through which accuracy may be improved.

2.4.2. Robustness treatments

Employers in our experiment who sample more from one group may do so for two reasons. First, because they hold more favourable beliefs to one group than the other. Second, because they have a desire to hold on to their prior beliefs about these groups; in order to achieve this, they may sample in such a way to avoid collecting information that may clash with their priors. To deal with the second motive, we designed the Probabilistic Exogeneity (PE) treatment.

Probabilistic Exogeneity (PE): Employers were not guaranteed to draw the shortlist that they requested. After making their shortlist selection, there was a 50% chance that they got a random shortlist of male and female workers drawn from all possible combinations other than the one chosen by the employer. Any of the alternative shortlist had an equal chance to be chosen by the computer. The PE treatment incentivises participants to truthfully reveal their shortlisting preferences, while allowing us to verify (for half the sample) how employers update their beliefs about information they would not have otherwise requested. Furthermore, we can see whether participants continue to sample in accordance with their beliefs, allowing us to control for people who sample in particular ways to hold certain desired beliefs.

To identify whether any perceived gender bias in belief updating is actually the result of a gender bias in belief updating, we need to ensure that it is not simply the result of updating beliefs about two different pools of workers. As such, we run a treatment in which there is no gender framing.

Only Beliefs – No Framing (OB-NF): This treatment was identical to OB-F, other than it was neutrally framed. We told employers that one pool was called “Pot-A” and the other pool is called “Pot-B”. There was no mention of gender in the instructions or in the information provided to employers during the experiment.

Additional treatment and balance test. In addition to the five treatments described above, we originally collected a sixth treatment, Only Beliefs – Shortlisting Samples (OB-SS). This treatment was designed to address a specific methodological concern: having collected unincentivized beliefs in the shortlisting treatments (such as BASE and SR) and incentivised beliefs in the beliefs-only treatments (such as OB-F), we aimed to test whether incentivisation affects belief updating. To do this, we created a beliefs-only treatment where the composition of the sample of ten workers mirrored the shortlisting patterns observed in the baseline treatment. A full description of the treatment is in Appendix A.2. However, an anonymous referee pointed out that the OB-SS treatment does not offer a clean comparison between incentivised and unincentivised belief updating. Unlike OB-F, the baseline treatment involved both belief elicitation and active shortlisting. As such, any comparison between OB-SS and BASE conflates the effects of incentivisation with differences in task structure.

Table 2
Means of control variables by treatment.

	Base	SR	PE	OB-F	OB-NF	OB-SS	F-Stat	p-val
1(Female Employer)	0.465	0.521	0.576	0.500	0.481	0.510	0.588	0.710
1(Watched Queen's Gambit)	0.366	0.340	0.394	0.310	0.311	0.410	0.775	0.568
1(Chess Rules?)	0.644	0.670	0.626	0.620	0.623	0.710	0.546	0.741
1(Play Chess Regularly)	1.693	1.819	1.788	1.740	1.660	1.810	0.648	0.663
1(Played Puzzle Rush)	0.030	0.032	0.020	0.080	0.028	0.090	2.053	0.070
1(Hiring Role)	0.505	0.468	0.475	0.460	0.604	0.450	1.360	0.238
Age	39.109	40.043	38.141	37.210	39.481	31.160	7.029	0.000
1(Student)	0.149	0.074	0.162	0.170	0.085	0.260	3.598	0.003
1(Full-Time Work)	0.416	0.489	0.444	0.380	0.509	0.440	0.916	0.470
Observations	101	94	99	100	106	100	.	.

The table displays the average of each characteristic for each group and the results of an F-test to check for differences in means across the treatments for each characteristic. All characteristics, except for age, represent the percentage of the sample possessing that characteristic.

While we could have still used elicited priors in OB-SS to boost our sample size, after collecting the OB-SS data, we conducted a balance test across all treatments. This revealed that randomisation failed on three participant characteristics. Specifically, age and student status differed significantly at the 0.01% level, and prior experience with the Puzzle Rush task differed at the 0.1% level. The results of this balance test can be found in Table 2. These imbalances were driven by the OB-SS treatment. We believe this was due to the staggered rollout of OB-SS relative to the other treatments. Because OB-SS required information about shortlisting behaviour in the baseline treatment, it could only be implemented after the main data collection was completed.

In light of these issues, we exclude OB-SS from the results reported in the main text. However, we report the originally planned analyses including OB-SS in Appendix A.3 for the sake of transparency. Importantly, our main findings are unchanged when OB-SS is excluded from the sample.

2.5. Design considerations

Social desirability and experimenter demand effects. A social desirability bias occurs if participants adjust their responses to align with what they perceive as socially acceptable or desirable. In our study, this could be a reduction in the difference in beliefs about performance across genders. Additionally, it could cause participants to not update their beliefs differently for each pool of workers.

An experimenter demand effect occurs when participants, aware that they are part of an experiment, alter their behaviour to align with what they believe the experimenters expect to find (Zizzo, 2010). In our context, if participants perceive that the experimenters are expecting to find a belief about a gender difference in performance or different updating behaviours for each pool, they may adjust their behaviours towards this expectation. Both biases are more of a concern when belief elicitation is unincentivised.

These two biases operate in opposite directions: social desirability bias might eliminate differences in priors and belief updating, while experimenter demand effects might induce participants to exaggerate these differences. To address some of these concerns, all the first round prior beliefs are incentivised. This enables us to get a closer look at our participants' true priors, which should mitigate some of these concerns about dampened or exaggerated priors about a difference in performance.

Generalised quadratic scoring rule. A limitation of the generalised quadratic scoring rule is that it is not robust to risk averse preferences. Risk averse participants may report a flatter belief distribution than that implied by their true beliefs. However, it preserves the rank order of the probability of each outcome. Given this, any beliefs about a difference in performance that we observe will be a lower bound on the size of the difference. Similarly, it is likely that the rate of change in accuracy is also a lower bound compared to the true rate of change that would take place.

3. Results

We now present our main results. We begin by investigating whether employers' priors are different for each gender, and whether gender impacts their belief updating in this context. We then study how beliefs impact shortlisting behaviour. Finally, we examine whether increased sampling from a group impacts the accuracy of beliefs.

3.1. Prior beliefs

We first examine whether there are gender differences in priors, both in terms of their means and their variances. To this effect, we use all round 1 data from all treatments except OB-NF. We collect the full belief distribution from each individual about ability in each pool, and from that information, we calculate the beliefs about both the mean and variance of ability.

Fig. 2 is a histogram of average beliefs about the performances of men and women in round 1. The x-axis shows the possible intervals for scores. The figure suggests there is a difference in distributions, primarily driven by the extremes: the belief distribution about men has more weight than that about women in the top two categories, while the reverse is true for the bottom two categories. A two-sample t-test confirms that the mean beliefs about the male pool are significantly larger than the mean beliefs about the female pool (Men: 16.64; Women: 16.20; $t = 2.78$; $p < 0.01$). However, there is no significant difference in the average belief about

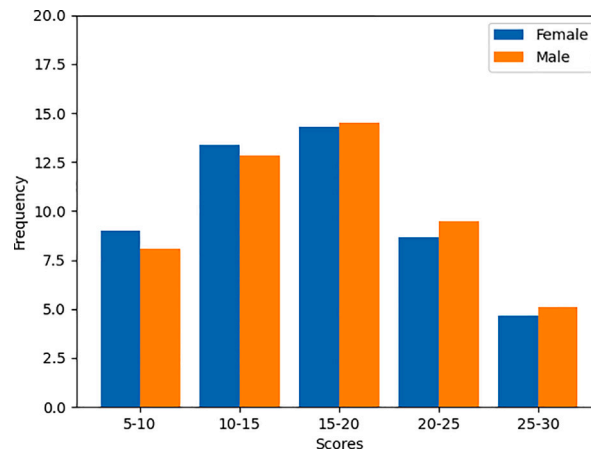


Fig. 2. Distribution of beliefs about men and women in round 1.

variance (Men: 32.15; Women: 32.25; $t = -0.12$; $p = 0.549$). OLS regressions (see Table 6 in Appendix A.3) confirm these results. The estimated difference in priors implies that women are believed to perform worse by roughly 3% on average.

Visual inspection of Fig. 2 suggests that the distribution of priors for males first-order stochastically dominates that of females. As exploratory analysis, we investigate this possibility.⁵ To this effect we check whether there are differences in the average belief about the number of men and women in each performance bin — for example, whether people perceive the number of individuals scoring 5–10, 10–15, and so on, to differ between the pool of men and the pool of women. We run a series of OLS regressions looking at the differences in means. We find that participants perceive there to be more women in the lower two bins and more men in the higher two bins, with no significant difference in the middle bin.

We note that the magnitude in the difference in beliefs (both mean and variance) is likely to be a lower bound on the true difference. First, the belief elicitation mechanism we used is based on the generalised quadratic scoring rule, which preserves the true relative ranking of probabilities but will generate probability distributions that are flatter than the true distributions for risk averse participants (Harrison et al., 2017). Second, our task removes the competitive element of chess through using chess puzzles; in doing so, we may also be removing a key driver of male stereotyping in chess, which could limit the observed beliefs about performance differences.

Result 1a: The mean of the distribution of prior beliefs about women are lower than that of men.

Result 1b: The variance of the distributions of prior beliefs about men and women are not different.

3.2. Mechanism 1: Differences in belief updating

Having established that there is a difference in priors about performance, we study two factors that may affect how beliefs are updated.

3.2.1. Gender-based differences in belief updating

We first verify whether or not our participants diverge from the reference Bayesian posterior mean when giving their own posterior beliefs (and by how much). In our setting, we were unable to elicit the full distribution of second-order priors as eliciting a probability for each possible combination for the numbers of workers who got each possible score is unfeasible to do in an experimental setting. In order to create a reference Bayesian posterior and to make this analysis possible, we assume that participants hold a set of prior beliefs of the possible compositions of the pots in accordance with a Dirichlet-Multinomial distribution. As the samples that participants draw have a discrete support, as expected under a multivariate hypergeometric distribution, we are able to update beliefs through the manipulation of the hyperparameters of the Dirichlet-Multinomial distribution.⁶ This allows us to generate our Bayesian benchmark in this situation where we cannot observe the entire distribution of priors.⁷

⁵ This and all exploratory analyses can be found in Appendix A.4.

⁶ The Dirichlet-Multinomial distribution is a family of probability distributions on finite, integer support. As such, it is well suited to our belief elicitation task, in which we elicit a probability distribution over five events from employers. Importantly, if the prior distribution of beliefs is Dirichlet-Multinomial and the likelihood function follows multivariate hypergeometric distribution, the posterior distribution of beliefs is also Dirichlet-Multinomial, which greatly facilitates our computations. This is due to the conjugacy of these prior and posterior distributions.

⁷ An important thing to note when calculating with the Dirichlet-Multinomial is that it is not able to handle a calculation where an interval is observed to be 0. In order to address this, we add a very small inconsequential observation to the interval of 0.0001. This allows us to compute a posterior when there are observations within this interval. Then, in order to tackle the potential that this means the posterior does not equal the prior when there are no observations for that population as a whole, the posterior is then reset to be the same as the prior, which would have been the case in a situation with no observations.

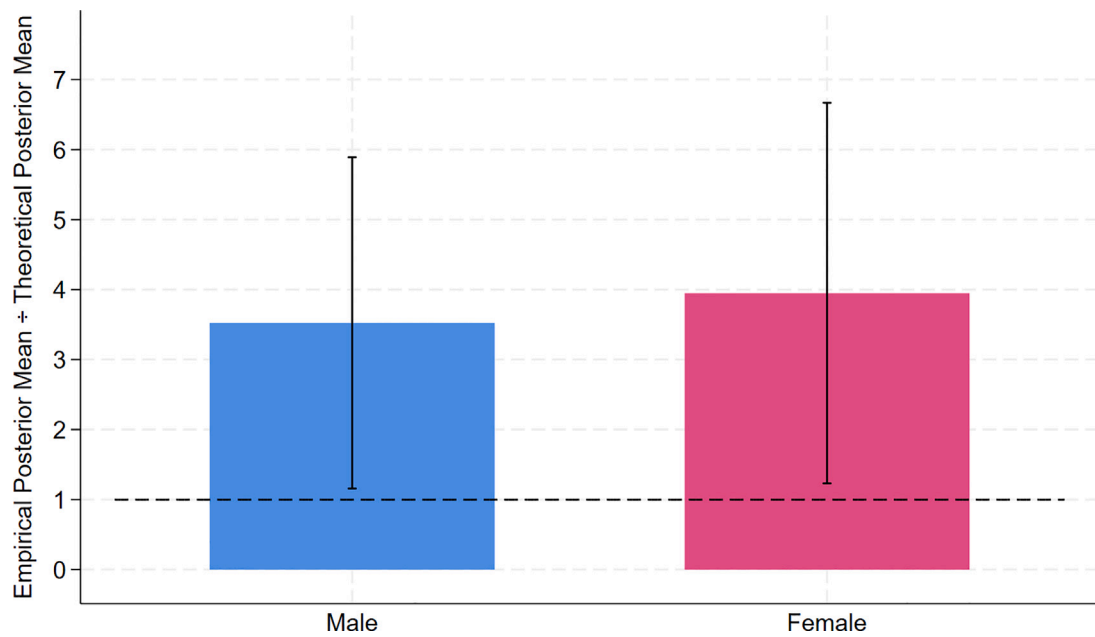


Fig. 3. Deviations in belief updating from the Bayesian benchmark.

Note: Bars represent marginal effects from a Poisson regression of the ratio of the empirical posterior mean to the Bayesian posterior mean on a gender pool dummy. Perfect Bayesian updating implies a ratio of 1. Whisker plots are 95% confidence intervals based on robust standard errors clustered at the individual level, recovered via the delta method.

After the theoretical distribution of priors have been calculated, we take the mean prior of the Dirichlet-Multinomial distribution to be the benchmark posterior distribution against which the participants' posteriors are compared. This benchmark allows us to calculate a new theoretical mean and compare it to the empirically observed mean of the participant's distribution. We do this by constructing a ratio for how the empirically observed mean diverges from the theoretical mean with the fraction $\frac{\Delta E_{i,g,t}}{\Delta T_{i,g,t}}$.⁸

In the full pre-registered analysis in Table 7 in Appendix A.3, we regress this divergence ratio on whether people are updating their beliefs about the male pool or female pool with controls for treatments. For expositional clarity, we present estimates from a non-preregistered version of this analysis using the OB-F treatment.

Fig. 3 presents the conditional mean estimates of the ratio of belief updating to Bayesian updating for each gender belief distribution. The left bar pertains to the female worker pool and the right bar pertains to the male pool. The average ratio is significantly larger than 1 for both pools, which indicates that employers generally over update compared to the Bayesian benchmark. However, employers do not update their beliefs differently for men and women. In the full analysis in Table 6 in Appendix A.3 for both the pre-registered version and previously presented non-pre-registered version, the coefficient on the Female Employer dummy is positive but not statistically significant.⁹ As such, we cannot reject the null that our participants updated their beliefs in the same direction as predicted by the Bayesian benchmark.

Result 2: *There is no statistically significant difference in how posterior beliefs deviate from the Bayesian benchmark posterior when updating beliefs about men and women.*

⁸ In order to deal with the issues surrounding a 0 theoretical mean change in the ratio $\frac{\Delta E_{i,g,t}}{\Delta T_{i,g,t}}$, for example in cases where they do not observe anyone from that group, we add 0.02 to the theoretical mean if for any observation the theoretical mean change is 0 and the difference between the empirical posterior and prior is positive; we subtract 0.02 from the theoretical mean if for any observation the theoretical mean change is 0 and the difference between the empirical posterior and prior is negative; and we code the ratio as 1 if the participant does not change beliefs and the theoretical mean change is 0. By doing so, we remove the possibility that this ratio is undefined, as well as giving participants the benefit of the doubt for that observation that they would have updated in the correct direction if the theoretical mean change had been anything other than 0. However, this does allow us to encode a "punishment" metric if they shift beliefs when they should not have. Additionally, to control for the highly influential effect of small theoretical means on this ratio, we choose to set the theoretical mean change to be 0.02 for any observation where the theoretical mean change is between 0 and 0.02; and we choose to set the theoretical mean change to be -0.02 for any observation where the theoretical mean change is between -0.02 and 0.

⁹ The noisy estimates could be due to a number of reasons. Some participants chose not to update their beliefs when the model predicted small changes in the distribution of beliefs. In the beliefs-only treatments, a theoretical change in mean smaller than 0.02 occurred 313 times, of which participants did not update beliefs 34 times, and a theoretical change in mean smaller than 0.1 occurred 583 times, of which participants did not update beliefs 123 times. Some participants updated their beliefs when it was not warranted, i.e., participants updated beliefs about a group when there were no observations about that group. This happened 43 times for beliefs about women and 54 times for beliefs about men. Instances of incorrect belief updating (i.e., in the wrong direction) occurred 584 times (20.53%).

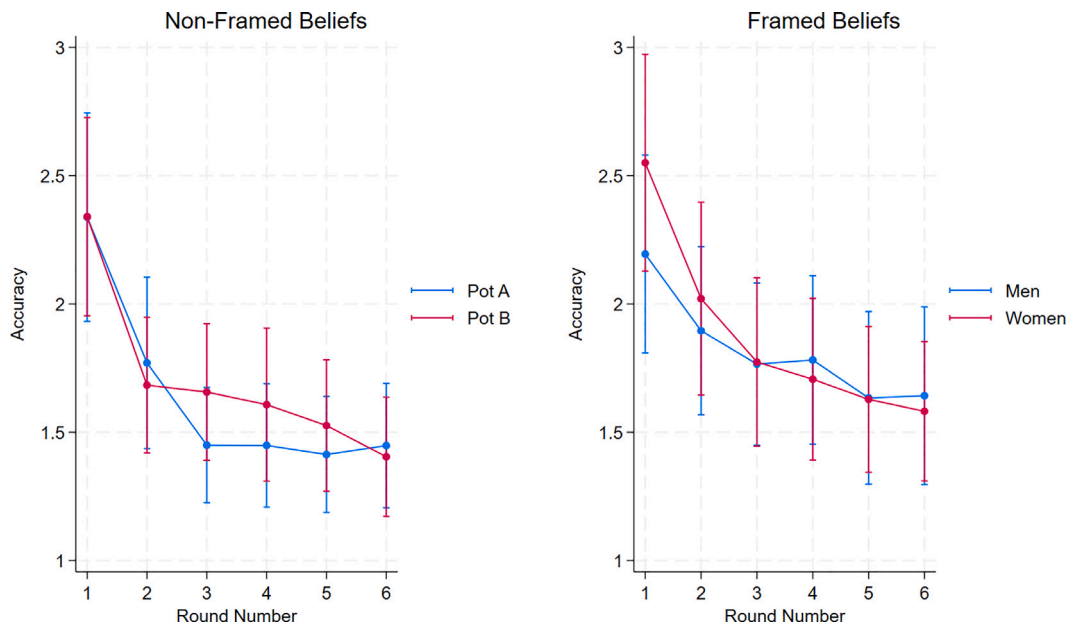


Fig. 4. Belief accuracy over time, OB-NF and OB-F treatments.

Note: Estimates of marginal effects from an OLS regression of the absolute difference between beliefs and performance on a gender pool dummy interacted with round dummies. Whisker plots are 95% confidence intervals based on robust standard errors clustered at the individual level, recovered via the delta method.

Robustness: We check how the framing of the pools impacted belief updating in the Only Beliefs treatments, by comparing belief updating in OB-F to that in the OB-NF treatment. In Table 7 in Appendix A.3, we provide estimates from a regression model in which we interact female worker with a OB-NF dummy. Using the estimates in model (2) in Table 7, we run a Wald test checking whether the addition of the coefficients of “1(Female Worker)” and “1(Female Worker) × OB-NF” is different from 0 ($F(1,305) = 0.13, p = 0.722$, FDR Sharpened q -value = 1). Under both the p -value and the FDR sharpened q -value, we fail to reject the null of a difference.

Additionally, we run three other checks on this result, detailed in Appendix A.5. First, we examine whether the result is robust when including only participants who shift their beliefs in the correct direction.¹⁰ Second, we test for the sensitivity of the divergence ratio to the choice of 0.02 as the adjustment factor. Finally, we assess divergence from the Bayesian benchmark using mean squared errors and Kullback–Leibler divergence. The result remains consistent across these analyses.

Exploratory: To provide an additional check on whether people are updating beliefs differently by gender in a more raw fashion,¹¹ we run an exploratory analysis in which we classify belief changes by whether they update in the correct direction given their observations, do not change beliefs or update in the wrong direction.¹² We find that people are not differentially likely to update in the correct or incorrect direction based on the gender of the worker pool. Additionally, we explore whether people update differently in response to shortlisting a “high-skilled worker” (whose score was between 25–30) or “low-skilled worker” (whose scores was between 5–10) and whether this differs by gender. We do not find strong evidence for this behaviour, with some very weak evidence that participants over-update less after they shortlisting a low-skilled woman.

3.3. Mechanism 2: Sampling behaviour

3.3.1. Does sampling more from a distribution lead to more accurate beliefs?

We consider next how the accuracy of beliefs evolves as employers sample more from each distribution. We initially consider the evolution of accuracy of beliefs in the beliefs-only treatments, since sampling is strictly exogenous. We then study the impact of the shortlisting restriction on the accuracy of beliefs in the shortlisting treatments. The measure of belief accuracy we use is the absolute distance of the mean of participants’ belief distributions from the mean of the true distributions of performance. A smaller absolute distance implies more accurate beliefs. Note that as we collect beliefs as an interval, we use the interval-midpoints for our mean calculation.

¹⁰ That is, participants shift the mean of their belief upwards if the theoretical posterior has a higher mean than their prior belief, and vice versa.

¹¹ As opposed to exclusively using the processed belief updating with the Dirichlet-Multinomial.

¹² If the mean of the observations the participants receive is higher than their subjective mean, they should move their beliefs about the mean upwards.

Table 3
Determinants of belief accuracy, BO-F treatment.

	(1)	(2)
Cumulative Observations	−0.027*** (0.006)	−0.057*** (0.015)
1(Second Half) = 1		−0.285 (0.214)
1(Second Half) = 1 × Cumulative Observations		0.040*** (0.014)
Constant	0.826 (0.527)	0.957* (0.517)
Controls	Yes	Yes
Observations	1200	1200
Clusters	200	200
R2	0.100	0.104

Estimates from Pooled OLS regression. The dependent variable in both models is the absolute distance between the mean of the participants' belief distribution and the mean of the true distribution of that group. Controls include dummy variables that equal 1 if: the employer is female; they have watched the Queen's Gambit; they know the rules of chess; they play chess regularly; they have played Puzzle Rush, the chess performance task; they have ever undertaken a hiring role at work, and 0 otherwise; and participant age. Robust standard errors clustered at the participant level in parentheses. * $p < 0.1$ ** $p < 0.05$, *** $p < 0.01$.

Fig. 4 presents the evolution over the course of the six rounds of the accuracy of beliefs about the performances of men and women in the framed and non-framed beliefs-only treatments. The downward trend in both treatments and both pools with each treatment indicates that the participants' beliefs become more accurate as they progress through the experiment.

We formally test the impact of having sampled from a population on accuracy of beliefs with a pooled OLS model; the estimates can be found in Table 8 in Appendix A.3. We display a shortened non-pre-registered version of this in Table 3 using the data from the OB-F treatment. In model (1), we present the regression of the accuracy of beliefs on the cumulative number of observations seen by participants, with controls. In model (2), we add a dummy variable that equals one for rounds 4–6 and an interaction between the two, with controls.

The coefficient on cumulative observations is negative and significant in both models, suggesting that belief accuracy about a group improves the more people an employer shortlists from that group. The coefficient on the interaction between cumulative observations and the last three rounds dummy is positive and significant, which suggests that the rate at which accuracy improves diminishes over the course of the experiment. This result is consistent with the non-linear pattern uncovered in Fig. 4. In other words, there may be some level of accuracy that can be easily reached and diminishing returns on information thereafter. Another possibility is that participants are becoming more familiar with the belief elicitation task; as a result, they become more precise in their estimates.

Exploratory: Classifying belief changes by whether they (i) update in the correct direction, (ii) do not change beliefs or (iii) update in the wrong direction given their observations, we find that employers are more likely to update in the correct direction if the theoretical shift in mean beliefs as given by our Bayesian benchmark is large. When participants beliefs are highly inaccurate, it is more likely that the size of the theoretical shift is larger, and as participants beliefs get more accurate, the size of the theoretical shift is smaller. This suggests that the rate at which a participant gets more accurate beliefs diminishes as the size of the theoretical shift decreases.

Result 3: Greater sampling from a distribution improves the accuracy of beliefs.

3.3.2. What is the relationship between subjective beliefs and sampling?

We now examine whether participants' shortlisting decisions were influenced by their beliefs about the ability of men and women, in terms of the mean and variance of their belief distribution. To this effect, we regress the number of women shortlisted on the difference in beliefs about the means of the male and female pools and on the difference in beliefs about the variance of the male and female pools, using a Poisson estimator. The full pre-registered results of this analysis are available in Appendix A.3. We display a shortened non-pre-registered version here using the data from the BASE and SR treatments in Table 4; we do not include data from the PE treatment as this is primarily a robustness treatment.

We find a positive and significant coefficient on Mean Diff (W-M) and a negative and significant coefficient on Var Diff (W-M). In other words, we find the expected positive association between mean beliefs and sampling, as well as the expected negative association between variance in beliefs and sampling. The coefficient on $SR \times Var\ Diff\ (W-M)$ is positive and significant, which indicates that the association between variance in beliefs and shortlisting is muted in the Shortlisting restrictions treatment. Column (2) reports marginal effects based on the estimated coefficients. An increase of the difference in perceived means by 1 puzzle is associated with an increase in the average number of women shortlisted of 0.210.

Table 4
Determinants of the number of shortlisted women, BASE and SR treatments.

	(1)	(2) Marginal effects
Mean Diff (W-M)	0.048*** (0.012)	0.210*** (0.038)
SR	0.025 (0.032)	0.143 (0.159)
SR × Mean Diff (W-M)	−0.012 (0.015)	
Variance Diff (W-M)	−0.008*** (0.003)	−0.020** (0.009)
SR × Var Diff (W-M)	0.007** (0.004)	
Constant	1.637*** (0.072)	
Controls	Yes	Yes
Treatment Interactions	Yes	Yes
Observations	1170	1170
Clusters	195	195
Pseudo-R2	0.016	

Estimates and marginal effects from Poisson regression. The dependent variable is the number of women shortlisted. Controls include dummy variables that equal 1 if: the employer is female; they have watched the Queen's Gambit; they know the rules of chess; they play chess regularly; they have played Puzzle Rush, the chess performance task; they have ever undertaken a hiring role at work, and 0 otherwise; and participant age. Robust standard errors clustered at the participant level in parentheses. Standard errors for marginal effects recovered through the delta method. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5
Effect of shortlisting restrictions on belief accuracy.

	(1)	(2)
SR	0.184 (0.131)	0.219 (0.145)
Second Half		−0.148* (0.082)
SR × Second Half		−0.070 (0.124)
Constant	1.855*** (0.284)	1.930*** (0.284)
Controls	Yes	Yes
Observations	2340	2340
Clusters	390	390
R2	0.013	0.017

Pooled OLS estimates, where dependent variable is the difference between beliefs about performance in a group and the actual performance in that group in BASE and SR treatments. Controls include dummy variables that equal 1 if: the employer is female; they have watched the Queen's Gambit; they know the rules of chess; they play chess regularly; they have played Puzzle Rush, the chess performance task; they have ever undertaken a hiring role at work, and 0 otherwise; and participant age. Robust standard errors clustered at the participant level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Result 4: *There is a positive association between shortlisting behaviour and the mean belief about performance. There is a negative association between shortlisting behaviour and the variance in beliefs about performance.*

3.4. Do quotas on shortlisting lead to more accurate beliefs?

We next examine whether imposing a quota on shortlisting from a group improves the accuracy of employers' beliefs about that group. To this effect, we regress the accuracy of the mean belief on a dummy for the SR treatment and its interaction with a dummy for rounds 4 to 6. We present the full pre-registered analysis of this in Table 10 in Appendix A.3. Table 5 summarises the estimates from a shortened version of this analysis using the BASE and SR treatments; we exclude the PE treatment from this analysis as it is primarily a robustness treatment.

Neither the coefficient on the impact of the shortlisting restrictions nor its interaction with the second-half dummy are statistically significant. As such, we cannot reject the null hypotheses that this form of restriction on shortlisting behaviour does not increase the accuracy of beliefs or that there is no differential impact of the shortlisting restriction over time.

Result 5: There is no evidence that shortlisting restrictions lead to more accurate beliefs. There is no evidence of a differential impact of the shortlisting restriction on belief accuracy over time.

Table 4 provides one reason for the lack of a significant effect of the SR treatment on shortlisting behaviour. Restricting shortlisting dampened the impact of the difference in variance between the two groups on shortlisting behaviour. In hindsight, it is likely that our shortlisting restriction was not strong enough to elicit a response. The mean number of women shortlisted in a round is 4.984 with a standard deviation of 1.901. In other words, the number of individuals for whom our pre-registered intervention could have had an effect were relatively small.

4. Discussion

Our first main question is whether there are differences in the priors of the distributions in our performance domain. Participants in our sample initially believed men to have a higher mean ability than women in solving chess puzzles, but we found no difference in the variance of their belief distribution. This discrepancy in beliefs across genders (in the chess domain) contributes to evidence on existing stereotypes in the difference in abilities between men and women, such as maths and sports (Carlane, 2019; Bordalo et al., 2019; Gupta, 2023).

There are a number of reasons why this initial stereotype exists. Participants may have been exposed to more high-performing men than women through the media. This is in line with a stereotype model of representativeness (Bordalo et al., 2016). In this case, the probability of seeing a highly rated male player is far higher than seeing a highly rated female player. Even though this may in part be a function of sample size, it is unlikely that people take this into account and as such may exhibit this type of representativeness bias (Kahneman and Tversky, 1972). It could also be that public statements made by experts such as Nigel Short guide public opinion, in line with the literature on how people take advice from experts (Yaniv, 2004). Finally, there is no objective data on performance of women and men in chess at the population level, due to the high self-selection of women away from competition (Smerdon, 2022). It is possible that males might be better at chess than females, and that the stereotype may be accurate. Understanding the underlying mechanism for stereotypes in chess is an important issue going forward.

When examining how participants updated their beliefs, we found that there is no difference in updating in terms of deviations from the mean of distribution of the theoretical mean posterior. In gender-stereotyped domains, this is a surprising result. When the literature has considered belief updating in stereotyped domains, they often find that participants are more likely to update in directions that allow them to hold these stereotyped beliefs (Bordalo et al., 2019; Gupta, 2023). These are usually considered to be indicative of motivated reasoning (Eil and Rao, 2011; Möbius et al., 2022; Thaler, 2020). In our experimental context, it could be that participants do not have strong desires to hold particular beliefs. When compared to a domain such as maths ability, chess ability might not be as closely tied to perceived norms. This suggests that participants do not gain as much utility from believing men are better in this chess domain (Akerlof and Kranton, 2000). This could be one reason as to why we do not see a difference in belief updating between two groups.

Our shortlisting restriction treatment had a very limited effect on behaviour. The main reason was the fact that the proportion of shortlisted women in the baseline treatment was high — the average shortlist had four women and six men. As a result, a minimum quota of one female turned out to be binding for only a very small number of cases. In labour markets where particular social groups are severely under-represented, it is possible that imposing shortlisting quotas may improve representation of minorities. The period following the implementation of the Rooney rule, which inspired the shortlisting restriction treatment, has coincided with a rise in non-white coaches, although the policy has been controversial (Grant, 2025). Closer to the specific domain of this paper, the introduction of gender quotas in team chess competitions has proven beneficial to female players, without any detrimental effects to male players (de Sousa and Niederle, 2022). More work to understand the impact of shortlisting quotas on representation is an important avenue for future work.

5. Conclusion

The literature on statistical discrimination has broadened recently to have a greater focus on the idea of inaccurate statistical discrimination (Bohren et al., 2019, 2020; Feld et al., 2022). Inaccurate beliefs can perpetuate the under-representation of a group in the labour market, because employers may be reluctant to even shortlist individuals from groups they perceive to be under-performing in some critical dimension (Lepage, 2023; Komiya and Noda, 2020; Gupta, 2023).

We run a stylised hiring experiment with a gender framing that investigates why people may persist in holding different beliefs about similarly performing groups. Stylised hiring tasks require some performance task against which hiring is undertaken. In our case, we use chess, specifically the performance on Chess.com's puzzle rush task. Throughout the experiment, participants are asked to construct a shortlist of 10 potential candidates from two groups, sampling however they wish from a pool of all-female players or a pool of all-male players, and then hire a candidate from this list. Additionally, we collect participants' prior and posterior beliefs about the full distribution of abilities of the workers in these two pools. While we choose to frame this through the lens of the labour market, it would be quite easy to extend this framing to a domain that is reminiscent of training young chess players. A chess coach has a limited amount of time and energy to go and find young chess players, and as such, if they have some prior that young men go on to be better at chess than young women, they may spend more time searching and training young men.

We find an initial gender-based difference in beliefs about the mean but not in beliefs about the variance. This finding about abilities in a chess domain further contributes to potential stereotypes in the difference in abilities between men and women, such as with maths and sports (Carlane, 2019; Bordalo et al., 2019; Gupta, 2023).

A key theme of this paper is understanding why inaccurate beliefs about similar groups may persist over time. To this end, we consider two mechanisms — one based on differences in belief updating and another based on sampling behaviour.

Considering the first mechanism, when compared to the Bayesian benchmark, we see that there is no difference in updating in terms of deviations from the mean of distribution of the theoretical mean posterior. In the context of the literature, this is a surprising result, as participants generally update to hold stereotyped beliefs (Bordalo et al., 2019; Gupta, 2023). Future research may wish to study the impact of the strength of the norm of a gender stereotype on gender differences in belief updating.

Regarding the second mechanism, we first consider how sampling impacts the beliefs about a distribution. We have evidence that both greater sampling from a group as well as greater time spent in sampling behaviour may generate more accurate beliefs. This contributes an important result to the literature on inaccurate belief discrimination. An important facet of inaccurate belief discrimination is considering how to correct inaccurate beliefs. The literature on the dynamics of belief discrimination has generally considered how these priors might be self-reinforcing (Lepage, 2023; Komiya and Noda, 2020). We contribute a route through which an employer might gain more accurate beliefs about populations.

To provide the second side of this mechanism, we consider how beliefs are associated with shortlisting behaviour. We find that both the subjective mean and subjective variance are strongly correlated with shortlisting behaviour in the expected direction. The result for the mean is inline with previous stylised hiring experiments that consider the mean of the distribution (Bohren et al., 2020; Coffman et al., 2021). Additionally, we contribute new experimental evidence of how the subjective variance is associated with hiring behaviour.

A key implication of our study is that encouraging contact with a group might help remove persistent inaccurate belief differences between groups. However, when implementing affirmative action policies, policy makers should ensure that it is strong enough to elicit the desired change.

Declaration of Generative AI and AI-assisted technologies in the writing process

No AI was used for the preparation of this manuscript.

Declaration of competing interest

The authors declare no competing interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jebo.2025.107115>.

Data availability

Upon publication, we will upload the data and replication file to this project's OSF page, where we have the project's preregistration.

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